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This Changes Everything: vLLM-Omni Unlocks True Multimodal AI for Developers

4 min read · Dec 4, 2025



Code Coup

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AI workloads are changing fast.

Modern inference pipelines no longer depend solely on autoregressive (AR) language models. Instead, they combine **different model types** — AR transformers for text, diffusion models for image and video generation, and specialized encoders for audio and multimodal understanding.

This shift requires a serving engine capable of efficiently handling **text, images, audio, video, and diffusion architectures**, without forcing engineers to juggle multiple frameworks.

Today, that finally becomes simple.

Introducing **vLLM-Omni** — an open-source framework that enhances the speed, usability, and cost efficiency of vLLM for *omni-modal* models such as **Qwen-Omni** and **Qwen-Image**.

If you already know how to use vLLM, you automatically know how to use vLLM-Omni.

vLLM-Omni

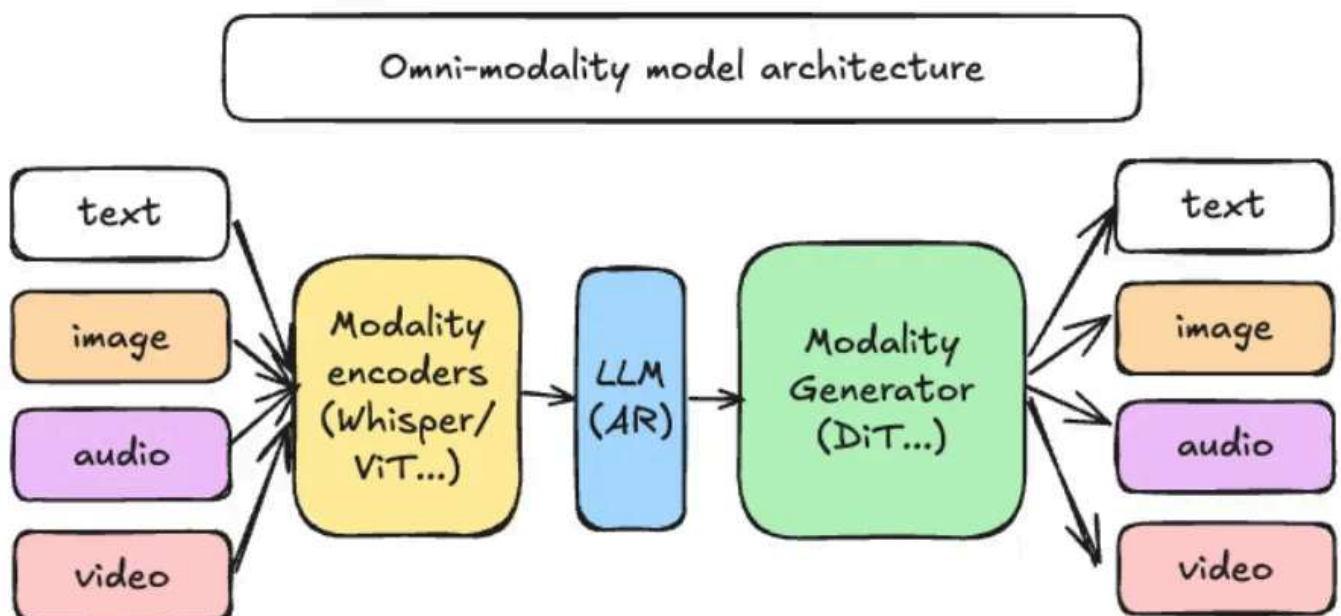
Easy, fast, and cheap omni-modality model serving for everyone

Why vLLM-Omni Matters

Traditional model serving frameworks primarily focus on text-based autoregressive models. Meanwhile, multimodal systems require:

- Image preprocessing
- Audio/video decoding
- Diffusion sampling
- Heterogeneous pipelines
- Parallel execution
- Dynamic scheduling across modules

vLLM-Omni brings all of this together in a **single, unified serving stack**.



Key Highlights

1. Full Omni-Modality Support

- Text
- Images
- Audio
- Video

2. AR + Diffusion + Parallel Generation

Run AR models (LLMs) and non-autoregressive architectures (DiTs, diffusion models) within a single pipeline.

3. Distributed & Efficient

- Pipeline parallelism
- Tensor parallelism
- Expert parallelism
- Dynamic resource allocation
- Streaming outputs
- OpenAI-compatible API

4. Works Out-of-the-Box with Hugging Face Models

Including:

- Qwen-Omni
- Qwen-Image
- And most multimodal open-source models

Getting Started with vLLM-Omni

Let's walk through the installation, setup, and execution of real multimodal inference.

Requirements

- **OS:** Linux
- **Python:** 3.12
- **GPU:** NVIDIA Compute Capability ≥ 7.0 (V100, T4, A100, H100, RTX 20xx, L4, etc.)

Note: vLLM-Omni is not natively supported on Windows.

Setup Using Python + `uv`

`uv` is a super-fast Python environment manager (think: `virtualenv` + `pip` on steroids).

Create a new environment

```
uv venv --python 3.12 --seed
source .venv/bin/activate
```

Install vLLM (base engine)

vLLM-Omni is built on top of vLLM v0.11.0.

```
uv pip install vllm==0.11.0 --torch-backend=auto
```

Install vLLM-Omni

```
uv pip install vllm-omni
```

That's it — you're ready.

Optional: Build from Source (if you want to modify vLLM-Omni)

```
git clone https://github.com/vllm-project/vllm-omni.git
cd vllm_omni
uv pip install -e .
```

If you're also modifying vLLM itself:

```
git clone https://github.com/vllm-project/vllm.git
cd vllm
git checkout v0.11.0
```

Set the wheel location:

```
export VLLM_PRECOMPILED_WHEEL_LOCATION=https://github.com/vllm-project/vllm/releases
```



Install:

```
uv pip install --editable .
```

Or, if PyTorch is already installed:

```
python use_existing_torch.py
uv pip install -r requirements/build.txt
uv pip install --no-build-isolation --editable .
```

Running Offline Inference (Qwen2.5-Omni)

The repository includes ready-to-run offline inference examples.

1. Running Multiple Prompts

First, download the `seed_tts` dataset and extract prompts:

```
tar -xf <path>/seedtts_testset.tar
cp seedtts_testset/en/meta.lst examples/offline_inference/qwen2_5_omni/meta.lst

python3 examples/offline_inference/qwen2_5_omni/extract_prompts.py \
  --input examples/offline_inference/qwen2_5_omni/meta.lst \
  --output examples/offline_inference/qwen2_5_omni/top100.txt \
  --topk 100
```

Navigate to the folder:

```
cd examples/offline_inference/qwen2_5_omni
```

Run inference:

```
bash run_multiple_prompts.sh
```

2. Running a Single Prompt

```
cd examples/offline_inference/qwen2_5_omni
bash run_single_prompt.sh
```

Using Local Media Files (Image, Audio, Video)

The `end2end.py` script supports multimodal input through CLI arguments.

Image-only inference

```
python end2end.py --query-type use_image --image-path /path/to/image.jpg
```

Video-only inference

```
python end2end.py --query-type use_video --video-path /path/to/video.mp4
```

Audio-only inference

```
python end2end.py --query-type use_audio --audio-path /path/to/audio.wav
```

Mixed modalities (audio + image + video)

```
python end2end.py --query-type mixed_modalities \
  --video-path /path/to/video.mp4 \
```

```
--image-path /path/to/image.jpg \  
--audio-path /path/to/audio.wav
```

Extract audio from video

```
python end2end.py --query-type use_audio_in_video --video-path /path/to/video.m
```



If you don't pass any files, default assets are used.

A Simple Python Example: Calling vLLM-Omni Programmatically

Here's what an ultra-simple multimodal inference request looks like:

```
from vllm_omni import OmniEngine  
  
engine = OmniEngine(model="Qwen/Qwen2.5-Omni")  
result = engine.generate({  
    "query": "Describe the image.",  
    "image": "/path/to/cat.png"  
})  
print(result["text"])
```

Or for mixed input:

```
result = engine.generate({  
    "query": "Summarize what is happening.",  
    "video": "/path/video.mp4",  
    "audio": "/path/audio.wav",  
})
```

librosa backend error?

Install ffmpeg:


```
sudo apt update  
sudo apt install ffmpeg
```

Final Thoughts

vLLM-Omni represents a major leap forward in model serving:

- **One engine** for text, image, audio, and video
- **One pipeline** for AR + diffusion
- **One interface** for multimodal AI

If you're building AI applications that involve any combination of text, images, audio, or video, vLLM-Omni is one of the simplest and most powerful ways to deploy them — open source, fast, and ready for production.

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Responses (1)



Eylsee

What are your thoughts?



Federico Cesarini

Dec 5, 2025



Wow this is great, I recently started self hosting with vLLM, and always struggled to work with multimodal context



Reply

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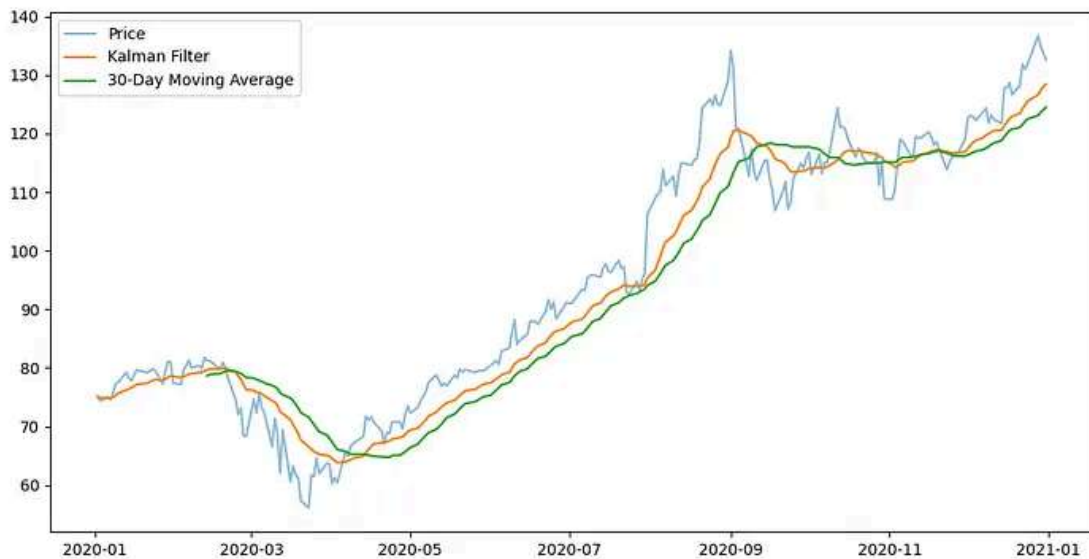


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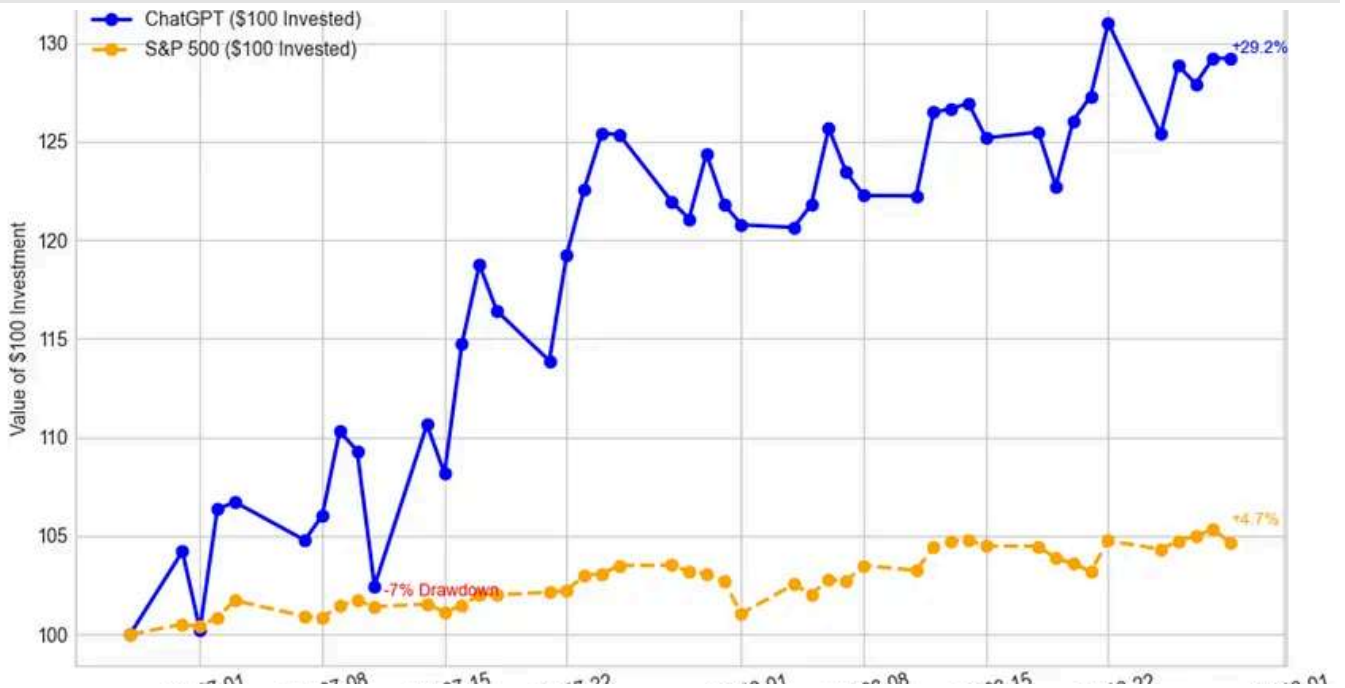
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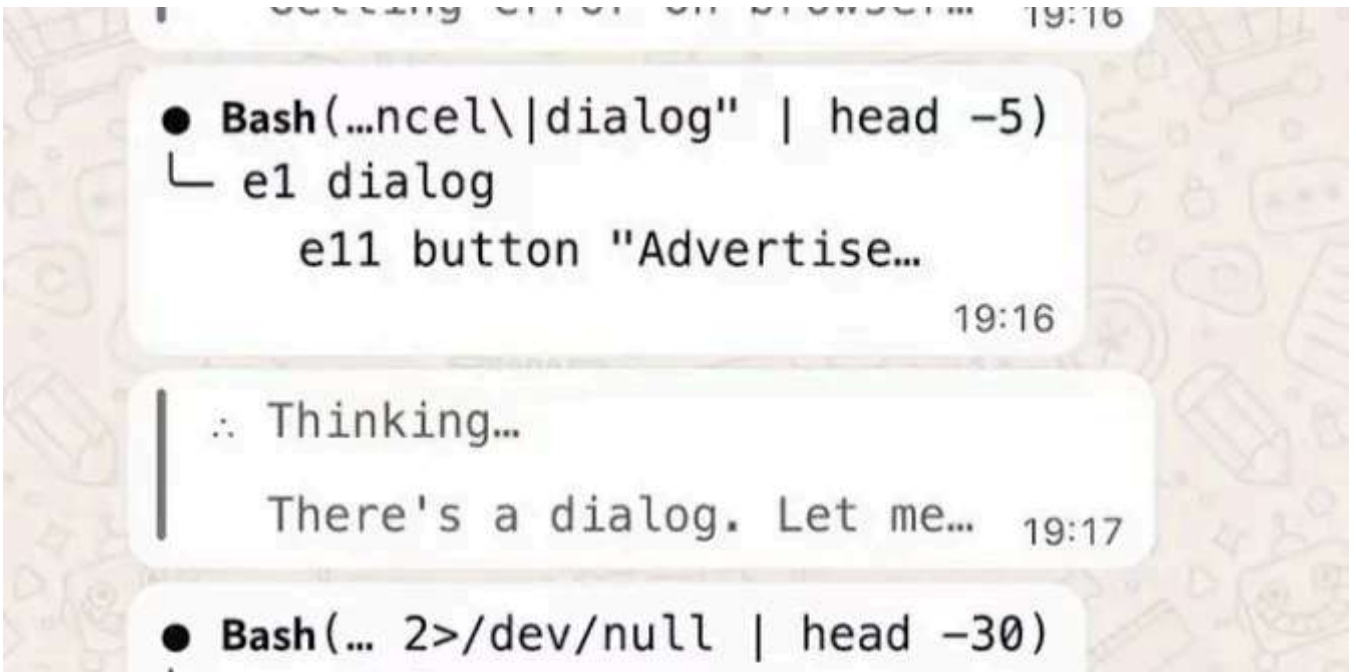
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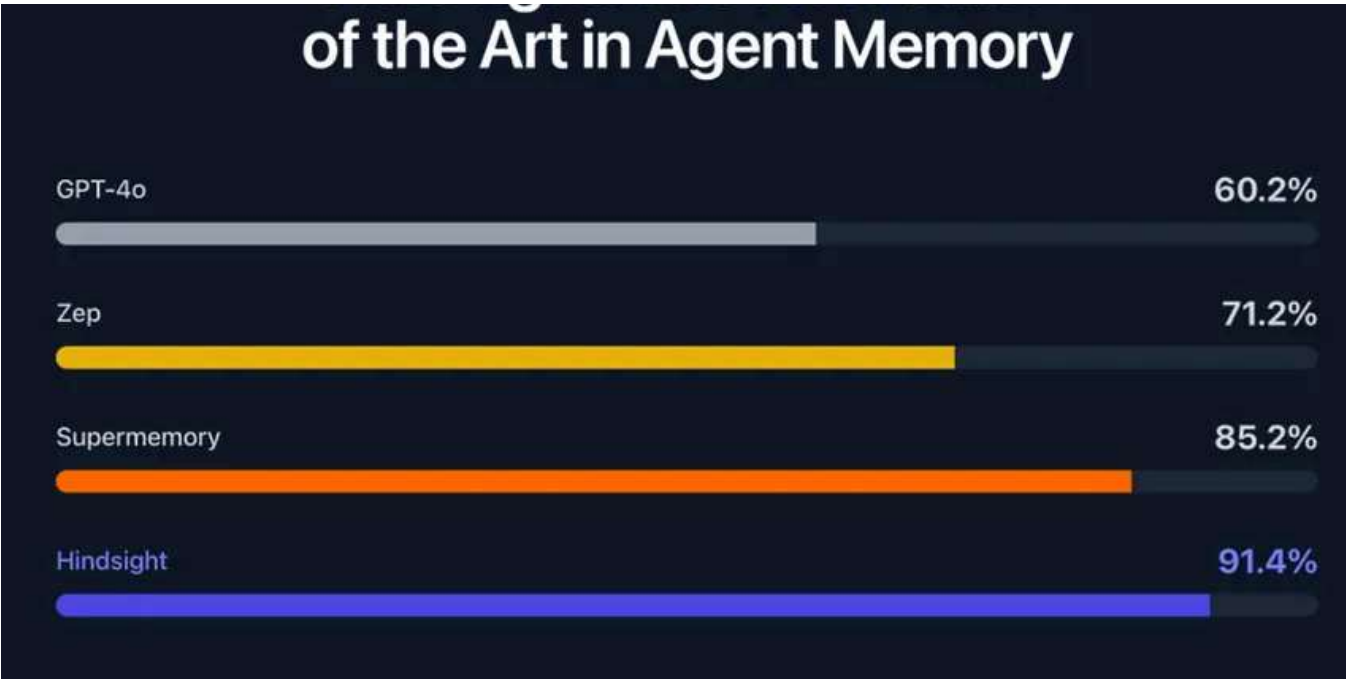
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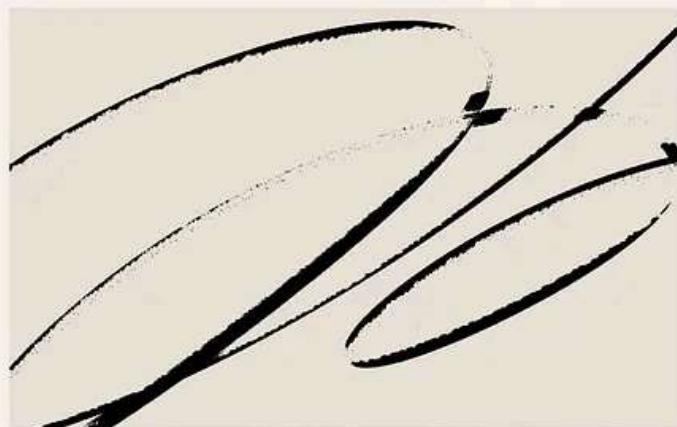
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	4	25	7	11	16	14	14
02_	5	1	3	5	1	3	3
	6	7	8	10	3	8	8
	7	10	11	22	7	2	9
-2_	8	3	5	3	5	7	5
02_	9	8	6	4	9	6	4
-2_	10	6	9	7	17	9	6

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